BSc Degree in Applied Statistics

Statistics in Health Sciences
1. Introduction

Jose Barrera[a b]
jose.barrera@uab.cat
https://jobago.github.io

[a]ISGlobal Barcelona Institute for Global Health - Campus MAR

[b]Department of Mathematics (UAB)

UAB Universitat Autònoma de Barcelona

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Statistics in Health Sciences

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Contextualization

Course information

About me

- **Contact:** Jose Barrera (jose.barrera@uab.cat)[a]
- **Training:**
  - PhD in Statistics (UPC, 2026)
  - MSc in Statistics (UB - UPC, 2012)
  - BSc in Statistics (UAB, 2007)
  - BSc in Physics (UB, 1994)
- **Positions:**
  - Biostatistician at ISGlobal[b] (2009 – present)
  - Associate Lecturer at Department of Mathematics, UAB (2008 – present)
  - External collaborator at Servei d'Estadística UAB (2012 – present)
  - Secondary school maths teacher (1995 – 2009 $\xrightarrow{\text{leave}}$ present)

[a] Further information: https://jobago.github.io

[b] Former Centre for Research in Environmental Epidemiology (CREAL)

Warning



During class sessions, **it is not allowed to:**

- Take photos,
- Record videos,
- Record audio.

Course information

Sessions

See details on theory and practice sessions schedule in the document **guia_SHS.pdf** (available at Moodle).

Evaluation

Activity		Scoring
Assignment 1 (in group)	t_1	$T = \frac{36\, t_{(1)} + 34\, t_{(2)} + 30\, t_{(3)}}{100}$ ($t_{(k)}$ is the k-th best score among t_1, t_2, t_3)
Assignment 2 (in group)	t_2	
Assignment 3 (in group)	t_3	
Exam (individual)	E	

- Course scoring: $Q = \begin{cases} \min\{T, E\}, & \text{si } T < 4 \text{ o } E < 3.5 \\ \frac{T+E}{2}, & \text{si } T \geqslant 4 \text{ y } E \geqslant 3.5 \end{cases}$
- For further details and doubts, read the document **guia_SHS.pdf**.

Course information

Assignments using reproducible research

- It is mandatory to write all the assignments under the **reproducible research philosophy**.
- Integration the usage of LaTeX and R in a single `Rnw` file using the R package **`knitr`**.
- Read carefully all details in the document `guia_SHS.pdf`.

Introduction to LaTeX& R with `knitr`

(For detailed guiding, see `setting_up_knitr.pdf`, in Exercise 0)

- **Installation (in that order):**
  1. Get R last version installed,
  2. Get a LaTeX editor (full installation is strongly recommended),
  3. Get RStudio last version installed.
- **Recommended setting up RStudio:**
  1. In the menu, go to Tools $\longrightarrow$ Global options... $\longrightarrow$ Sweave,
  2. Set "Weave Rnw files using:" to "knitr",
  3. Set "Typeset LaTeX into PDF using:" to "pdfLaTeX".
- **Practice:**
  1. Exercise 0 will be developed at classroom on.
  2. Exercise 0 includes an `Rnw` template that can be used to write the assignments.

Biostatistics is statistics applied in fields related to biology. Some examples:

Public health

Multidisciplinary field (sociology, economics, biology, ...). The main aim is to maintain and improve the global health, as well as to prevent, control and eradicate diseases. We will focus on:

- **Epidemiology**, which is related to the distribution, frequency, causes and control of the factors related to health and diseases in the population.

Genomics

Focuses in the operating, structure, evolution and origin of genomes. Avant-garde field. Related to molecular biology, biochemistry, informatics, statistics, mathematics and physics.

Ecology

Related to geostatistics, point processes and lattice data.

Clinical trials

Experimentally assess the effectiveness and safety of a given product, substance, medicine, diagnosis or therapeutic technique.

Barcelona Biomedical Research Park (PRBB)



Barcelona Biomedical Research Park (PRBB)



Barcelona Biomedical Research Park (PRBB)

Description

- Initiative of the Catalan Government, the Barcelona Council and Universitat Pompeu Fabra
- One of the main biomedical research centres at international level
- Linked to Hospital del Mar
- $\approx$ 1400 people
- $\approx$ 45 different countries
- $\approx$ 30% of the scientific staff are foreigners (communication is often in English)
- $\approx$ 50% are under 36 years old
- $\approx$ 60% are women
- Academic collaborations (BSc, MSc, and PhD) in universities (UPF, UAB, UB, UPC, ...)

Barcelona Biomedical Research Park (PRBB)

Research centres

At PRBB there are 7 research centres ($\approx$ 100 groups), related to:

- Bioinformatics, epigenetics, cellular biology, pharmacology, human genetics, etc
- . . . and also to Epidemiology $\longrightarrow$ Environmental epidemiology $\longrightarrow$ ISGlobal - Campus MAR

ISGlobal-MAR

Aim

ISGlobal-MAR
(former Centre for Research in Environmental Epidemiology (CREAL))

"To identify the environmental agents related to health
and to promote the prevention and control."

ISGlobal-MAR

Some research programmes

Programme	How it affects ...	to ...
Respiratory	Environmental, working and genetic factors	Respiratory diseases (asthma and COPD*)
Cancer	Environmental and genetic factors	Bladder, breast, colon, leukemia, lymphoma,...
Child health	Particulate matter (PM_x), organochlorides,...	Development, reproduction, neuroconductual system
Air pollution	, and others	Cardiorespiratory and neurodevelopment systems
Water pollution	Desinfectants for drinking water and pools	Cancer, respiratory diseases, reproductive disorders, ...
Radiation	Ionizing ( ) and non-ionizing ( )	Tumours and others

*COPD: Chronic obstructive pulmonary disease.

Contents

PubMed: Search of scientific papers. Structure of a paper

Searching literature & publishing

- PubMed
  - What is PubMed?
  - How does it work?
- Structure of a paper

Classification of studies

How studies are designed?

- Notation
  - What is the exposure?
  - What is the outcome?
- Classification criteria: what characteristics of the study define its classification?
- Types of epidemiological studies
- Advantages and disadvantages of each study design

Classification of variables and related regression models

Variables and modeling: statistics vs epidemiology

- According to the measure type
- According the role in the study
- Types of explanatory variables
- Types of regression models according to the metric of the response variable
- Response variables of type time

Dealing with missing data

In the real world, data are always imperfect...

- Dealing with missing data
- Multiple imputation

Example of statistical methods in Health Sciences: Integration of multiple imputation in cluster analysis

Combining known methods to solve new problems...

- Working paper: "A framework for multiple imputation in cluster analysis" [1]
- Software: The R package miclust

Measures of disease presence

How can we measure health indicators?

- How many individuals are affected by a given virus at a given time?
- What is the probability of suffering a given disease within a given time interval?
- What is the occurrence rate of a given disease?

Measures of association between exposure and disease

How can we relate health indicators and exposures?

- How can we measure the association between suffering a given disease and having been exposed to a given condition or circumstance?
- How can we compare the risk of suffering a given disease in the presence and absence of a given exposure?
- What indicators can we consider? Are they always estimable?

Causality, confusion and interaction

Association does not imply causation...

- A positive correlation between milk consumption and cancer implies a causal relationship?
- What aspects should we consider when deciding whether exists a significant association between a given exposure and a given disease?

Example of statistical methods in Health Sciences: Regression models with transformed variables. Interpretation and software

Helping investigators (and statisticians) to interpret logarithmic transformations...

- Working paper: "Models with transformed variables: interpretation and software" [2]
- Software: The R package tlm

Optional lectures

Further reading

- **Statistical modeling:**
  Agresti[3], Clayton and Hills[4], Hosmer and Lemeshow[5], McCullagh and Nelder[6].
- **Epidemiology:**
  Gordis[7], Rothman et al.[8], Rothman and Lash[9].
- **Cancer research:**
  Breslow and Day[10], Breslow and Day[11], Breslow and Day[12].

References

[1] X. Basagaña, J. Barrera-Gómez, M. Benet, JM. Antó, and J. Garcia-Aymerich. A Framework for Multiple Imputation in Cluster Analysis. *American Journal of Epidemiology*, 177(7):718–725, 2013. URL https://doi.org/10.1093/aje/kws289.

[2] J. Barrera-Gómez and X. Basagaña. Models with transformed variables: interpretation and software. *Epidemiology*, 26(2):e16–17, 2015. URL https://doi.org/10.1097/EDE.0000000000000247.

[3] Alan Agresti. *Categorical Data Analysis (3rd Edition)*. John Wiley & Sons, Inc., Hoboken, New Jersey, USA, 2013.

[4] D. Clayton and M. Hills. *Statistical Models in Epidemiology*. Oxford University Press, Oxford, UK, 1993.

[5] DW. Hosmer and S. Lemeshow. *Applied Logistic Regression (3rd Edition)*. John Wiley & Sons, Inc., Hoboken, New Jersey, USA, 2013.

[6] P. McCullagh and JA. Nelder. *Generalized Linear Models (2nd Edition)*. Number 37 in Monographs on Statistics and Applied Probability. Chapman & Hall/CRC, Boca Raton, Florida, USA, 1989.

[7] L. Gordis. *Epidemiology (5th Edition)*. Elsevier, Amsterdam, The Netherlands, 2013.

[8] KJ. Rothman, S. Greenland, and TL. Lash. *Modern Epidemiology (3rd Edition)*. Wolters Kluwer/Lippincott Williams & Wilkins, Philadelphia, Pensilvania, USA, 2008.

[9] KJ. Rothman and TL. Lash. *Modern Epidemiology (4th Edition)*. Wolters Kluwer/Lippincott Williams & Wilkins, Philadelphia, Pensilvania, USA, 2020.

[10] NE. Breslow and NE. Day. *Statistical Methods in Cancer Research Volume I: The Analysis of Case-Control Studies*. Number 32 in IARC Scientific Publication. International Agency for Research on Cancer, Lyon, France, 1980.

[11] NE. Breslow and NE. Day. *Statistical Methods in Cancer Research Volume II: The Design and Analysis of Cohort Studies*. Number 82 in IARC Scientific Publication. International Agency for Research on Cancer, Lyon, France, 1987.

[12] NE. Breslow and NE. Day. *Cancer Epidemiology: Principles and Methods*. Other Non-Series Publications. International Agency for Research on Cancer, Lyon, France, 1999.